# A Dual-Tank Model for Real-Time Tracking of Phosphocreatine and Glycolytic Energy Reserves in Cycling

**White paper — AnaerobicFuelTanks project** *Version 0.5 · 2026-07-11 (fourth revision — oxidative-headroom-gated PCr recovery; τ_g → 470; both-bars-affine scope narrowing; first empirical results on real rides)*

> **Scope and status (read this once).** This is a **physiologically-motivated decomposition of one measured quantity — single-tank W′bal — whose split fraction `f_p` is assumed, not measured** (power cannot identify the depletion split; §4.2). The two-bar display is precisely characterised, and the characterisation is narrow: **during steady supra-CP effort both bars are affine in W′bal** ( `R_p/C_p ≈ R_g/C_g ≈ W′bal + const` ; §4.2, §4.4), and when the PCr tank is full — most of the time — the PCr bar is likewise an affine transform of W′bal. **All non-redundant content lives in exactly two transients:** (i) the glycolytic *activation ramp* at effort onset, governed by `τ_on` (measured) and `τ_off` (**not** measured), and (ii) the *PCr recovery transient* after hard efforts, governed by `τ_p` and its new oxidative-headroom gate (§4.2). **There is still no experiment showing the two-bar decomposition is *more correct* than single-tank W′bal** — that requires the recovery-law head-to-head (§6.6), and at the compartment level it is unfalsifiable in-modality *even in principle* (§6.5, §8). But v0.5 adds the first **usefulness** evidence from real rides (§6.8): on interval sessions the PCr bar is informative (below 95%) ~45% of the time and the depletion caps move it up to ~25–60 pts vs a recovery-only null at the moments that matter. `f_p` , `τ_off` , and the emergent `τ_dep` remain assumed; `τ_g` /reconstitution were verified against primary sources (§4.1a). The rest of the paper builds on these; it does not re-litigate them.

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## Abstract

Endurance-cycling analytics have converged on the Critical Power (CP) / W′ framework, in which all work performed above a sustainable power asymptote is drawn from a single finite reserve, W′. This lumps two physiologically distinct anaerobic systems — the fast phosphocreatine (PCr / alactic) system and the slower glycolytic (lactic) system — into one tank with a single recovery time constant. That simplification is convenient but discards the information a rider most wants during hard, variable-intensity efforts: *which* reserve is being spent, and *how fast* each will come back. This paper proposes a **dual-tank model** that splits W′ into a fast PCr reserve and a slow glycolytic reserve, drives both from the instantaneous power trace, and applies system-specific depletion and restoration laws drawn from the bioenergetic and 31P-MRS literature.

We are explicit up front about what this is: a **physiologically-motivated decomposition of one measured quantity (W′bal) whose split is assumed, not measured**, and whose extra resolution over single-tank W′bal is real but concentrated in the transients after hard efforts. It generalizes W′bal in depletion and diverges from it — by design — in recovery. The model is deliberately reduced to be computable at 1 Hz on a wrist- or bar-mounted device, and we specify a Garmin Connect IQ implementation that displays live reserve, consumption, and depletion for both systems. We close with a validation strategy — including the one out-of-sample test that would actually earn the second tank — and an explicit statement of the model's assumptions and limits.

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## 1. Introduction

A power meter reports mechanical output, not metabolic cost. The bridge between the two is a model of the body's energy-supply systems. The dominant bridge in cycling — the CP/W′ model — treats supra-CP work as depletion of a single reservoir and sub-CP work as its exponential refill (Skiba et al. 2012). It is power-native, validated, and computationally trivial, which is why it underlies W′bal displays in GoldenCheetah, intervals.icu, and several head units.

Its central simplification is also its central weakness for real-time decision-making: **W′ is one tank.** Physiologically, anaerobic capacity is at least two tanks with very different dynamics (Margaria; di Prampero & Ferretti 1999):

- The **phosphocreatine (PCr / alactic)** system — high power, low capacity, **fast** oxidative recovery (seconds to a minute), but with recovery that is pH-sensitive and slows across repeated bouts.
- The **glycolytic (lactic)** system — larger capacity, lower peak power, and **slow** recovery (minutes) that effectively only proceeds at low intensity.

A rider deciding whether to cover the next surge cares which tank is empty. If the PCr tank is full but glycolytic is depleted, a 5-second jump is fine but a 40-second dig is not — a distinction the single-tank W′bal cannot express. The goal of this paper is a model that keeps the power-native, on-device virtues of W′bal while restoring the two-system resolution.

Two honest qualifications, stated here rather than buried in §7. First, the split is a **modeling assumption**, not a measurement — power alone cannot identify how W′ divides between the two systems (§4.2). Second, the extra resolution is **concentrated in transients**: whenever the fast PCr tank is full — which, given its ~27 s recovery, is most of the time — the glycolytic reserve is just an affine rescaling of the existing single W′bal, so the second bar adds decision-relevant information mainly in the ~30–60 s after a hard effort (which is, however, exactly when covering-the-next-attack decisions are made). The model earns its keep as a *heuristic decomposition*, not as two independently measured reserves.

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## 2. Physiological background

**Phosphocreatine (alactic).** ATP is buffered by the creatine-kinase reaction (PCr + ADP → ATP + Cr). Intramuscular PCr (~20–25 mmol·kg$^{-1}$ wet muscle) is the highest-power energy source and is drawn on instantly at any power transient. Its resynthesis is oxidative and fast: classic alactic $O_2$-debt half-time ≈ 30 s (Margaria); 31P-MRS gives recovery time constants τ_PCr ≈ 20–40 s, strongly **pH-dependent** (acidosis slows it) and **biphasic** under high glycolytic load (Harris et al. 1976; Yoshida et al. 2013, PMID 23662804). Recovery also **slows progressively across repeated hard bouts.**

**Glycolytic (lactic).** Anaerobic glycolysis converts glycogen to lactate, supplying large ATP flux for ~10 s–2 min. Its "fuel level" is best tracked as accumulated muscle/blood lactate — a proxy for $H^+$ and metabolite accumulation. Contribution decays across repeated sprints (≈40% of ATP in the first of 10×6 s efforts, <10% by the last; Sci Rep 2024, DOI 10.1038/s41598-024-78916-z). Restoration is slow (lactic $O_2$-debt half-time ≈ 15 min) and, mechanistically, proceeds fastest at low intensity, effectively ceasing above the first lactate threshold (LT1).

**The onset that matters.** Both systems contribute from the start of a hard effort, but not equally at first: PCr is drawn on instantly (it is the highest-power, immediate buffer), while glycolytic flux takes several seconds to ramp up as glycogen phosphorylase is activated (Parolin et al. 1999) — so the *early* demand is PCr-weighted, shifting toward a shared draw as glycolysis engages. This graded onset — and the very different refill rates — is what the dual-tank model must reproduce (it does so with a parallel, PCr-weighted draw and a glycolytic activation ramp; §4.2), not a strict "PCr first, then glycolytic" hand-off.

**What the two "tanks" represent — and what they don't.** The "tank" is a deliberate simplification for intuition and on-device display, not a claim that either system is a literal reservoir that drains. The two sides are not even the same *kind* of quantity:

- The **PCr tank** does correspond to a real, depletable substrate store — intramuscular phosphocreatine — so "PCr empty" is close to literal.
- The **glycolytic "tank"** is better understood as **tolerance to accumulating fatigue-related metabolites** than as a fuel gauge. On the ~1–4 min timescale it represents, muscle glycogen is nowhere near exhausted; carbohydrate depletion is a separate, hours-long limiter this model does not address. What forces power down at the limit of a hard effort is the buildup of inorganic phosphate (Pi), $H^+$ (acidosis), ADP, and extracellular $K^+$ toward the limit of tolerance — consistent with exhaustion at CP/W′ coinciding with a reproducible low-PCr / high-metabolite state. (Lactate itself is largely a fuel and a marker, not the cause of force loss.)

This is the physiological reason the two recovery laws differ: PCr resynthesizes quickly, whereas the glycolytic side recovers slowly and only at low intensity because it reflects *clearance and buffering* of accumulated byproducts, which need aerobic metabolism and time — exactly why the model tracks the glycolytic state as an accumulation variable (a muscle-lactate proxy; §4.1) and gates its recovery below LT1.

Muscle fatigue is multifactorial and still debated; W′ deliberately lumps it into one number for usability. In short: **the PCr tank is a fuel that depletes; the glycolytic "tank" is a byproduct bucket that fills.**

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## 3. Prior models (and why a reduced dual-tank sits between them)

Three model families exist (full survey in `literature-review-anaerobic-models.md`):

- **Family A — CP / W′-balance** (Skiba 2012/2015; Bartram 2018; Froncioni–Clarke). Power-native and on-device-friendly, but **one lumped tank, single recovery τ.** Recovery `τ_W′ = 546·e^(−0.01·D_CP) + 316 s` cannot represent fast PCr + slow glycolytic simultaneously.
- **Family B — hydraulic tank models** (Morton 1986; Weigend `three_comp_hyd` 2021). **Two explicit anaerobic vessels** (AnF = phosphagen, AnS = glycolytic) plus an aerobic vessel; out-predicts W′bal for intermittent recovery (arXiv 2108.04510). But its 8 parameters are abstract and fit per-athlete by evolutionary computation — heavy for an embedded target.
- **Family C — bioenergetic supply/demand ODEs** (EJAP 2023, PMID 37369795). Explicit alactic / lactic / aerobic metabolic-rate terms, with an efficiency-limited PCr recovery and a power-gated lactate-removal law. Mechanistically ideal but has 17 parameters and needs grey-box fitting.

**The gap:** Family A is implementable but under-resolved; Families B and C resolve the two systems but are too heavy to run and calibrate on a head unit. The dual-tank model below keeps the two-tank resolution of B/C with the on-device simplicity of A.

To be precise about what "lighter" means here, because it is easy to oversell: the model's core free-parameter count (`CP, W′, f_p, P_p_max, g_rate, τ_p, τ_g, τ_on, LT1`, with `τ_off = τ_on` by default — **nine free**, plus two optional realism terms) is essentially the hydraulic model's eight — **the same size, not a smaller model.** The difference is *where the parameters come from*. Only `CP` and `W′` are fitted per athlete (from a standard CP test); the rest are **hard-coded to literature defaults** rather than fit by evolutionary computation. That buys on-device simplicity and a trivial calibration, at a real cost: the load-bearing defaults (`f_p`, `P_p_max`, `LT1`, and the recovery τ's) are exactly the quantities a CP test does **not** determine, so the model's two-system resolution rests on assumptions, not measurements. That is a legitimate identifiability-for-convenience trade — but it is a trade, and the sections below are explicit about which numbers are earned and which are assumed.

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## 4. The dual-tank model

### 4.1 State and parameters

Two energy reserves (Joules), each with a fixed capacity:

- `R_p` — **PCr reserve**, capacity `C_p`
- `R_g` — **glycolytic reserve**, capacity `C_g`
- Total anaerobic capacity `W′ = C_p + C_g`

The model has **nine free parameters** (plus `τ_off = τ_on` and two optional realism terms). Only `CP` and `W′` come from a CP test; the rest are literature-fixed defaults (see §3 — this is a real count, not a small one):

| Symbol | Meaning | Typical default |
|---|---|---|
| `CP` | critical power (W) | from 3-/12-min CP test (≈ FTP + a few %) |
| `W′` | total work above CP (J) | from same test (~15–25 kJ) |
| `f_p` | fraction of W′ assigned to the PCr tank | **0.25** — *assumed* (physiology, not data; §7) |
| `P_p_max` | PCr peak power above CP (W), at a **full** tank | ≈ $P_{1s}$ − CP (best 1 s power − CP) |
| `g_rate` | glycolytic peak flux as a fraction of PCr peak flux (a ratio) | 0.5 → `g_pmax = g_rate·P_p_max` (flux-ratio note) |
| `τ_p` | PCr recovery time constant (s) | **27** — within the 31P-MRS `τ_PCr` range (20–40 s), gated by oxidative headroom (§4.2) |
| `τ_g` | glycolytic recovery time constant (s) | **470** — reconstitution fit (Ferguson 2010 at 20 W ⇒ gate ≈ 0.90; joint optimum with `f_p`; §4.1a) |
| `τ_on` | glycolytic activation time constant (s) | 6 (Parolin 1999) |
| `τ_off` | glycolytic **de**-activation time constant (s) | = `τ_on` — *assumed* (Parolin measured activation, not de-activation; §7) |
| `LT1` | first lactate threshold power (W); glycolytic tank recovers below it | **measured** (a threshold test), *not* a fixed %CP |

Derived: `C_p = f_p·W′`, `C_g = (1 − f_p)·W′`. Initial state `R_p = C_p`, `R_g = C_g` (full).

**Free-parameter count.** Nine free (`CP, W′, f_p, P_p_max, g_rate, τ_p, τ_g, τ_on, LT1`); `τ_off = τ_on` by default. Only `CP` and `W′` are fitted per athlete; the rest are literature-set. That is essentially the hydraulic model's eight — **the same size, but the parameters are literature-set rather than fitted per athlete** (the real trade; §3). A PCr *depletion* kinetic, `τ_dep = C_p/P_p_max`, also falls out of the above (§4.4) and is equally assumed — see §7.

**`η` was removed (was 0.80).** The former "recovery efficiency" only rescaled the effective PCr recovery constant to `τ_p/η` — no sub-`C_p` asymptote, so not a true efficiency, and degenerate with `τ_p` when fitted. It silently made the effective constant ≈ 27.6 s while the table advertised 22 s. **`τ_p = 27 s` is now justified on its own terms:** it sits inside the 31P-MRS `τ_PCr` range of 20–40 s (Yoshida 2013; Harris 1976) — *not* as "22 inflated by η" (that would just re-import the double-count). The `eta` settings key remains, defaulting to 1.0 (identity) with a **deprecation note**, so a future maintainer cannot silently reintroduce the rescale. (The pulmonary-$VO_2$ recovery half-time of 74 s in Ferguson 2010 is *slower* than muscle `τ_PCr`, because it includes circulatory processes — see the component-mismatch discussion in §4.1a.)

The `g_pmax = 0.5·P_p_max` ratio is not arbitrary: peak glycolytic ATP flux is roughly half peak PCr/creatine-kinase flux in human muscle (di Prampero & Ferretti 1999; the biopsy time-courses of Parolin 1999 / Bogdanis 1996), so PCr is the higher-power system by about 2:1.

Several of these deserve flags up front, because the defaults are doing real work:

- **`f_p` is assumed on physiological grounds, but now *corroborated* by the recovery data (§4.1a).** We set **0.25** from physiological alactic-fraction estimates (the PCr system supplies ~20–30% of anaerobic ATP capacity; di Prampero & Ferretti 1999, Bangsbo 1990), and — once the reconstitution reference protocol's recovery power is known (20 W, near-passive) — the reconstitution curve independently implies `f_p ≈ 0.20–0.25`, converging with the physiology. Treat it as a per-athlete calibration target; the supported band is ~0.20–0.25 (a joint reconstitution fit centres on 0.20).
- **`P_p_max ≈ P₁ₛ − CP` is the rate at a FULL tank.** PCr flux is not constant: the rate *ceiling* tapers with tank fullness (creatine-kinase equilibrium — flux falls as PCr depletes), so the available PCr rate is `P_p_max·(R_p/C_p)`, equal to `P_p_max` only when the tank is full (§4.2). Read `P₁ₛ − CP` as an upper bound: at 1 s glycolysis is already ~15% active, so it mildly over-attributes to PCr. Note this taper is on the *ceiling* only — it does **not** set the submaximal share (§4.2).

#### 4.1a Reconstitution: the source, the calibration, and a contradiction the source raises

The 37% / 65% / 86% W′-recovery at 2 / 6 / 15 min (half-time 234 s) is **Ferguson et al. 2010** (*J Appl Physiol*, DOI 10.1152/japplphysiol.91425.2008), reviewed by Chorley & Lamb 2020, who state it was measured "when recovering at a **nominal 20 W**." Three things follow.

- **`τ_g` is calibrated to ~470 s, not confounded (but not exactly "passive").** The confound of review 3 — the LT1 gate makes the effective constant `τ_g/gate`, so the fit depends on recovery power — is resolved by the 20 W figure. But 20 W is not quite `gate = 1`: with `LT1 ≈ 0.8·CP ≈ 200 W`, `gate = (200−20)/200 = 0.90`, so a model reproducing Ferguson needs `τ_g ≈ 470 s` (± a few % on the cohort's unreported LT1), **not 520**. The default is corrected to **470**.
- **`f_p` is *consistent with* the curve, not independently corroborated by it.** With recovery power fixed, the curve still constrains a **ridge in `(f_p, τ_g)`**, not `f_p` alone: `(f_p 0.25, τ_g 520)` and the true joint optimum **`(f_p 0.20, τ_g 470)` (SSE ≈ 10.7 pts²)** both fit 37/65/86 to within ~3–5 pts. So the honest statement is *one* physiological estimate of `f_p` (di Prampero & Ferretti 1999, alactic $O_2$-debt t½ ≈ 30 s; Bangsbo 1990, ~20% alactic) **plus a reconstitution curve consistent with it** — the physiology is what selects `f_p ≈ 0.25` on the ridge. We keep `f_p = 0.25` as the physiological midpoint but note the recovery data alone would prefer ~0.20. *(An earlier draft called this "two independent lines converge"; that overstated the independence — corrected here.)*
- **Ferguson's own components contradict the model's identification — and this is the strongest single datum in the paper.** Ferguson reports the two recovery channels underlying W′ reconstitution:

$VO_2$ t½ = 74 s (a proxy for oxidative/PCr resynthesis) and blood-lactate t½ = 1366 s. Put those beside the model's component half-times:

| Component | Model | Ferguson channel | ratio |
|---|---|---|---|
| fast | `τ_p = 27 s` → t½ **18.7 s** | $VO_2$ **74 s** | 4.0× slower |
| slow | `τ_g = 470 s` → t½ **326 s** | lactate **1366 s** | 4.2× slower |

**Both of the model's components are ~4× too fast, yet the aggregate 234 s half-time reproduces.** That is the signature of fitting two exponentials to a *sum* without checking them against the *components* — the identifiability problem this review series has circled, now with a number. And Ferguson's own conclusion is blunter still: W′ reconstitution is "**not a unique function of phosphocreatine concentration or arterial [lactate], and it is unlikely to simply reflect a finite energy store that becomes depleted.**" *The primary source used to calibrate the recovery law explicitly denies that the recovery decomposes the way the model decomposes it.* We host that here rather than resolve it by citation-picking. Two defences, and their limits: (i) $VO_2$ off-kinetics is **not** PCr resynthesis — it includes slower circulatory processes — and muscle 31P-MRS `τ_PCr` of 20–40 s does support `τ_p = 27`; so `τ_p` is defensible against the 31P literature while being 4× off against the channel measured in the very protocol that produced the recovery curve. (ii) The channel *separation* (74 s vs 1366 s, ~18×) genuinely supports a bi-exponential structure, even if the channel *values* are not the tanks. The tension is the finding; the two-tank recovery is a heuristic that fits the aggregate, not a measured decomposition of it.

## 4.2 The 1 Hz update

Let `P` be instantaneous power (W) and `Δt = 1 s`. Define the demand relative to aerobic supply as `Δ = P − CP`.

**Case 1 — depletion (`Δ > 0`, above CP).** Demand `need = Δ·Δt` (J) is met by **both systems in parallel**. This matters: glycolytic ATP supply is not instantaneous. Glycogen phosphorylase (the rate-limiting glycolytic enzyme) transforms to its active form over the first several seconds of maximal effort — Parolin *et al.* (1999) measured it rising from 12% at rest to 47% by 6 s — while phosphocreatine is the immediate buffer (a fresh 30 s sprint drains it to ~17%; a 10 s sprint leaves ~20–30%; Bogdanis *et al.* 1996); both contribute from the onset, not in sequence (González-Alonso *et al.* 2000). We model this with a **glycolytic activation** term `g ∈ [0,1]` that ramps first-order with a time constant `τ_on ≈ 6 s` while above CP and relaxes back during recovery:

```
g ← g + (1 − g)·(1 − exp(−Δt/τ_on))       # glycolytic activation ("inertia")

# --- SHARE of submaximal demand: capacity-proportional (NOT rate-proportional) ---
w_p = C_p                                  # at g=0 → PCr covers all; at g=1 → PCr cov
w_g = C_g · g
share_p = need · w_p / (w_p + w_g)         # (guard: if w_p+w_g ≈ 0, share_p = need)
share_g = need − share_p

# --- RATE CEILING: peak flux, PCr tapered with fullness. Governs MAXIMAL efforts only
rate_p = P_p_max · (R_p / C_p)             # (guard: C_p > 0)
rate_g = g_rate · P_p_max · g

take_p = min(share_p, R_p, rate_p·Δt)      # each tank capacity- AND rate-limited
take_g = min(share_g, R_g, rate_g·Δt)
unmet  = need − take_p − take_g
# shortfall (a tank empty or rate-capped) spills to the partner, then to a deficit:
take_g += min(unmet, R_g − take_g, rate_g·Δt − take_g);   unmet -= …
take_p += min(unmet, R_p − take_p, rate_p·Δt − take_p);   unmet -= …
R_p -= take_p;  R_g -= take_g
D  += unmet                                # DEFICIT (debt): supra-CP work the caps co

if unmet > 0:  rate_limited = true         # producing power beyond the tanks' flux —
```

**Why the share and the ceiling are different objects** (this is the fix that ended a three-round patch cycle). The rate ceiling — `P_p_max`, `g_pmax` — encodes "PCr is the higher-power system." That is true at **maximal** effort, where both systems are flux-limited, and it is where PCr dominance belongs. It is the **wrong** object for apportioning **submaximal** supra-CP demand, where neither system is near its flux limit and the sharing is set by metabolic control, not peak-flux ratios. Earlier revisions used one rate-weighted rule for both jobs, which forced the PCr trajectory to be a fixed, convex, intensity-*invariant* function of W′-spent (PCr at ~9% by the midpoint of *any* hard effort). Weighting the **share by capacity** and keeping the taper on the **ceiling** fixes it with no cost in the sprint case (there the caps and spill dominate and the share rule never binds — verified bit-identical):

- **Submaximal:** both tanks drain ≈ proportionally to capacity, so `R_p/C_p ≈ R_g/C_g ≈ W′bal` and both reach their nadir *together* at exhaustion — matching the reproducible-metabolic-milieu picture in §2. Honestly, this means during a steady supra-CP effort the two bars track W′bal and each other; they diverge only in the activation ramp and in recovery. That is not a defect — it is the model behaving the way the Scope box and §7 already say it does (extra resolution lives in transients).
- **Maximal:** the tapered ceiling makes `R_p` decay ~geometrically to its nadir at exhaustion (a ~10 s all-out sprint leaves `R_p ≈ 20–30%`, matching Bogdanis 1996; §4.4), and PCr dominance emerges from

the ceiling, not the weight.

- **Energy conserved even when a cap binds.** Residual `unmet` is banked in a **deficit** `D` (standard W′bal permits a negative balance), so `(R_p + R_g − D)` drops by exactly `Δ`·Δt per second and §6.2 holds. `D` clears only below LT1 (Case 2) — it is supra-cap byproduct load and must respect the same intensity gate as the glycolytic tank. Note the deficit preserves the *aggregate* but not the *split*: work booked to `D` never drains a tank, so if `D` is large the two bars read slightly full. `D` is an accounting term for the rare supra-flux case, usually signalling a **stale** `P₁ₛ`, not fatigue.

**On identifiability (and consistency with §7).** The parallel draw is the physiologically faithful choice for the *live display*, and `τ_on` is a literature-set constant (Parolin 1999), not a fitted one. But we are careful not to over-claim: the depletion split is **not** identifiable from power at all — total draw above CP is all the pedals see, and any (`f_p`, `g_pmax`) that sums to the same total is observationally equivalent. In *principle* the bi-exponential **recovery** of W′ constrains the split (a fast PCr component + a slow glycolytic component); in *practice* that fit is ill-conditioned (§6.3 shows the fast component is invisible at standard reconstitution sampling), so `f_p` ends up **assumed and weakly constrained**, not recovered. This is the §7 position, and it is the one to trust — the reserves are a plausible decomposition, not measured latent state.

**Case 2 — restoration (`Δ ≤ 0`, at/below CP).** Recovery "headroom" is `CP − P`. Each tank refills exponentially toward full; glycolytic refill is gated by intensity.

```
g ← g · exp(−Δt/τ_off)                            # glycolytic DE-activation (τ_off =

# PCr: fast, oxidative — resynthesis needs aerobic ATP ABOVE the ride's own demand, so
# gated by the OXIDATIVE HEADROOM (CP − P): full at rest, near-arrested at CP.
gate_p = max(0, (CP − P) / CP)
R_p += gate_p · (C_p − R_p) · (1 − exp(−Δt/τ_p))  # τ_p from 31P-MRS τ_PCr (§4.1)

# Glycolytic tank AND the deficit clear only below LT1 (a MEASURED power, not a %CP)
if P < LT1:
    gate = (LT1 − P) / LT1                        # 0 at LT1, → 1 toward zero power
    R_g += gate · (C_g − R_g) · (1 − exp(−Δt/τ_g))
    D   -= gate · D · (1 − exp(−Δt/τ_g))          # debt clears on the same gate as R_
```

**The PCr recovery gate is new in v0.5, and it fixes the model's headline use case.** Until now PCr recovered at the same rate whether the rider was stopped or holding 99% of CP — so the "punch" bar went green while the rider was still under load, in exactly the "*can I cover this next attack?*" scenario the field exists for (sitting in a bunch between attacks is recovery at high sub-CP power). PCr resynthesis is oxidative and needs spare aerobic capacity, which vanishes as P → CP; the `gate_p = (CP − P)/CP` term encodes that. At P = 0 it is identical to the old law (`gate_p = 1`), so nothing in depletion or the reconstitution fit moves; at tempo it does the physiologically obvious thing. Verified: after a 20 s / 700 W effort (PCr → 23%), 60 s of recovery now restores PCr to **87% at 0 W, 46% at 0.8·CP, and 25% at 0.99·CP** — versus a flat 92% at every recovery power before the gate.

The `g` de-activation is what makes the activation ramp re-fire on each interval of a repeated-bout set (without it, `g` would ratchet to 1 on the first surge and every later bout would start at a flat split — the ramp inert for the rest of the ride). **`τ_off = τ_on` is an assumption, and a load-bearing one:** Parolin (1999) measured phosphorylase *activation*, not *de*-activation, which is a separate quantity. If de-activation runs on a minutes timescale rather than seconds, the ramp would not re-fire on 30 s recoveries and the repeated-sprint behaviour changes materially — so `τ_off` is flagged alongside `f_p`, `τ_g` (§7).

`LT1` is the rider's **first lactate threshold in watts**, from a threshold test — *not* a fixed fraction of CP. LT1 and CP vary independently (LT1 is roughly 65–85% of CP depending on training status), and this gate decides whether the glycolytic tank refills at all during tempo/endurance riding, so getting it wrong there is consequential. Where only CP is known, `LT1 ≈ 0.80·CP` is a fallback, but it is the weakest default in the model and should be replaced by a real measurement.

**Optional realism (recommended, off by default): -** *pH-slowed / fatiguing PCr recovery:* scale `τ_p` upward as the glycolytic tank is emptier, e.g. `τ_p_eff = τ_p · (1 + k·(1 − R_g/C_g))`, capturing the observed slowing across repeated bouts. **-** *Aerobic ramp:* replace the hard `CP` boundary with a first-order aerobic supply `A(t)` that rises toward `min(P, CP)` with `τ_aer ≈ 25 s`, and use `need = (P − A)·Δt`. This reproduces the onset "oxygen deficit" that transiently loads the anaerobic tanks even below CP.

### 4.3 Reported quantities

- **PCr reserve** `= R_p / C_p` (0–100%) — how much "punch" is left.
- **Glycolytic reserve** `= R_g / C_g` (0–100%) — how much "sustained dig" is left.
- **Consumption rate** (per system, W) — `take_p/Δt`, `take_g/Δt`, the live draw on each tank.
- **Combined W′bal** `= (R_p + R_g − D)/W′` — the deficit term keeps this **energy-conserving in depletion** (it drops by exactly `Δ`·Δt per second) so it matches a single-tank W′bal reference on above-CP segments. It intentionally **diverges in recovery** (bi-exponential + LT1 gate; §4.4/§6.2), so it is *depletion-compatible*, not identical everywhere.
- **Depletion / exhaustion flag** — the rider is producing power beyond the tanks' rate/capacity (`D` growing), or both reserves are at/near zero.

### 4.4 Why this behaves correctly

- A short, very hard surge draws mostly from `R_p` (glycolysis has not yet ramped in), and `R_p` refills after — *at a rate set by recovery intensity* (§4.2 gate). A maximal ~10 s sprint leaves **`R_p ≈ 20–30%`, not empty**, from the tapered ceiling's emergent depletion constant `τ_dep = C_p/P_p_max`. **But `τ_dep` is not a fixed 7 s** — §7 notes it swings ~4–13 s across riders, so the 10 s residual ranges from ~10% (fast riders) to **~46%** (`τ_dep ≈ 13 s`). The central case matches Bogdanis 1996 (PCr ≈ 17%

at 30 s); the high tail does *not*, which is an argument for a *measured* PCr depletion constant (the re-architect fork, §7), not an emergent ratio.

- A sustained supra-CP effort draws from **both** tanks, declining together to their nadir at exhaustion. The drawdown is **front-loaded (mildly concave), not linear** — the activation ramp makes the early draw PCr-weighted (450 W: 68/42/18/1 at quarter-TTE intervals). During such steady efforts **both bars are affine in W′bal** (§4.2), so they carry no information single-tank W′bal lacks *there*; their only non-redundant content is the ramp offset (set by `τ_on` / `τ_off`) and the post-effort PCr recovery transient. This is stated as a limitation, not a feature.
- **A falsification test (not a validation), and its real cost.** For a constant supra-CP effort, `R_p/C_p` is *asymptotically* a function of the fraction `φ` of W′ spent — but only for efforts long relative to `τ_on` and below the ceiling regime; the activation ramp adds a genuine second intensity dependence (a hard effort reaches a given `φ` faster, with `g` lower, so PCr is *lower* at that `φ`). So the test "hold different supra-CP powers to exhaustion and compare PCr-at-%W′-spent" would report intensity-*dependence* for a correct model, and a naive read would wrongly "falsify" it. Framed correctly it is a **falsification** test (pass ⇒ the bar is a deterministic restatement of W′bal in depletion, i.e. redundant; fail ⇒ the architecture is wrong) — lose–lose for the *depletion* claim, which is the point. And it is **not** "testable in an afternoon": checking PCr in-cycling needs 31P-MRS, which §6.5 says is unavailable in this modality. It requires a surrogate (post-exercise sampling, or knee-extension 31P-MRS with a cross-modality transfer assumption; §8).
- **Synthetic battery (v0.5).** Parameters: `CP = 255 W`, `W′ = 20 kJ`, `f_p = 0.25`, `P_p_max = 690 W` (`g_rate = 0.5`), `τ_p = 27`, `τ_g = 470`, `τ_on = τ_off = 6`, `LT1 = 204 W`.

| Test | Result |
|---|---|
| Sustained 450 W — PCr at 25/50/75/100% of TTE | 68 / 42 / 18 / ~1% (front-loaded, nadir at exhaustion) |
| Both tanks at exhaustion | PCr ~1% / GLY ~0% (empty together) |
| 945 W (= $P_{1s}$) 10 s sprint — PCr residual | 23% (ceiling-dominated; share rule doesn't bind) |
| 6×[10 s@700 W / 30 s@150 W] — PCr at each bout end | 49→17→10→8→8→7% (gated recovery: 150 W is high sub-CP, so little refill) |
| 1200 W hold (caps bind) — combined W′bal conservation | leak = 0 J |
| PCr recovery after 700 W/20 s, 60 s @ {0, 128, 204, 242, 252} W | 87 / 68 / 46 / 30 / 25% (headroom gate) |
| Debt repayment at 0.9·CP (above LT1) | `D` unchanged (LT1-gated) |

- **On "generalizing W′bal" — a careful claim.** In *depletion* the model is a faithful generalization: `(R_p + R_g − D)` falls by exactly `Δ`·Δt per second (cap-binding or not), so the summed draw above CP matches standard W′bal. In *recovery* it is deliberately **different** (bi-exponential, LT1-gated), so it does not reduce to Skiba's single `τ_W′`. It is a generalization in **structure and depletion**

**behaviour**, not a strict superset. (We drop the earlier `f_p → 0` argument: with the fullness taper `C_p = f_p·W′ = 0` is a division by zero, guarded in code — the limit is not meaningfully computable and does not add anything.)

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## 5. Real-time on-device implementation (Garmin Connect IQ)

The model's per-second cost is a handful of multiplies and one `exp()` per tank — trivial for a head unit. A companion document, `connectiq-app-spec-and-prompt.md`, gives the full technical specification and a ready-to-use build prompt. Summary of the target design:

- **App type:** a custom full-screen **Data Field** ( `Toybox.WatchUi.DataField` ), which the head unit calls once per second — a natural fit for the `Δt = 1 s` update.
- **Input:** `Activity.Info.currentPower` inside `compute(info)` .
- **Configuration:** `CP` , `W′` , `f_p` , **`P_p_max`** (a per-athlete calibration target — §6.7), **`LT1`** (a measured threshold, not a %CP default — §4.2), and the recovery constants exposed via Connect IQ app settings ( `Application.Properties` ), so a rider enters them from Garmin Connect.
- **Display:** two vertical bar gauges (PCr and glycolytic reserve) with numeric % and color bands (green → amber → red). *Per-system live consumption in W is a "modelled share," not a reading* — when the ceiling is slack it is just `(power − CP)` split by a fixed fraction (§7), so it is labelled as such or omitted.
- **Recording:** write both reserves and consumption to the FIT file via `FitContributor` fields so they sync to Garmin Connect / intervals.icu / Strava for post-ride analysis.
- **Footprint:** a handful of scalar state variables ( `R_p` , `R_g` , the activation `g` , the deficit `D` ), no arrays — well within Connect IQ memory budgets. Per-second cost is still a few multiplies and one `exp()` per tank.

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## 6. Validation strategy

The model is a hypothesis; it must be checked against data before its numbers are trusted. The honest difficulty is that the model's *headline feature* — a separate PCr reserve tracked in real cycling — is the hardest thing to validate, so we separate tests that only check the plumbing from the one test that would actually earn the second tank.

1. **Face validity / unit tests** — synthetic power traces (single sprint, repeated sprints, supra-CP hold, sub-CP recovery) should produce the qualitative behaviours in §4.4.
2. **Backward compatibility — *depletion only*.** The combined balance `(R_p + R_g − D)` reproduces a reference W′bal implementation (Froncioni–Clarke) on the same trace **in depletion** — it drops by exactly `Δ` ·Δt per second, and the deficit term `D` preserves that even when a rate cap binds. It will **not**

match in recovery, by construction (§4.4), so this test must be scoped to above-CP segments — a test that expected agreement everywhere is one the model is designed to fail.

3. **Reconstitution targets — what the curve constrains, and what it cannot.** Combined recovery after a full W′ depletion is compared to the published curve (37 / 65 / 86% at 2 / 6 / 15 min; Ferguson et al. 2010, at 20 W recovery). (a) **The curve is blind to the fast tank** — a `τ_p ≈ 27 s` process is `1 − e^(−120/27) ≈ 99%` complete by the first (2 min) sample, so it validates *glycolytic* recovery and the *size* of the PCr offset, **not** PCr kinetics (for those you need 10/20/30/45/60 s samples; test 4). (b) **The curve fixes `τ_g` and constrains a `(f_p, τ_g)` ridge, not `f_p` alone.** With the 20 W recovery power known (gate ≈ 0.90), the fit gives `τ_g ≈ 470` and the joint optimum `(f_p 0.20, τ_g 470)`; `f_p` is *selected on that ridge by physiology*, not by the curve (§4.1a). So the curve is **consistent with** `f_p ≈ 0.20–0.25`, not an independent measurement of it. *(This supersedes a v0.3 statement — since deleted — that computed an implied `f_p ≈ 0.13` at the old `τ_g = 360 s`; that value was a fossil of the superseded default.)*
4. **Fast-recovery kinetics (the missing piece).** The load-bearing test for `τ_p` is a repeated-bout protocol with **early** recovery sampling — e.g. all-out efforts separated by 10/20/30/45/60 s — and the between-bout power recovery correlated with modelled PCr recovery (as Bogdanis 1996 did for PCr). Standard reconstitution data cannot supply this; it must be measured deliberately.
5. **System-specific truth (lab) — and its hard limit.** Blood-lactate kinetics can corroborate the glycolytic tank, but the PCr tank's gold standard, **31P-MRS, is essentially unavailable in real cycling**: the measurement requires the muscle to be inside a magnet, so it exists only for small-muscle knee-extension or immediate post-exercise, not whole-body cycling. The model's headline novelty therefore **cannot be checked against the gold standard in the modality it targets** — a limitation to state plainly, not paper over.
6. **The test that earns the second tank — and what it actually tests.** Because the model is a near-superset of single-tank in depletion, "it fits better" is guaranteed and is **not** evidence the compartments are real. The decisive test is a **head-to-head against single-tank W′bal on an out-of-sample intermittent-tolerance prediction** (the protocol class where the hydraulic model beat W′bal, arXiv 2108.04510). But be precise about what it probes: since this model's *depletion* is single-tank by construction, its **only** lever to beat single-tank is the **recovery law** (bi-exponential + LT1 gate). So **test 6 is a test of the recovery law, not of the two-compartment hypothesis** — the framing "two tanks → better prediction" overstates what a win would show. This also sharpens the protocol: choose interval recovery-valley powers that **straddle LT1**, where the dual-tank and single-tank recovery laws diverge most, rather than a generic interval set.
7. **Field calibration — scoped honestly.** Fit **`P_p_max` and the `τ`'s** per athlete to maximal and repeated-bout tests. **`f_p` is *not* a routine calibration target:** §4.2 shows the depletion split is unidentifiable from power, and only the test-4 class (all-out efforts with early recovery sampling) can constrain it — and even then the fit is ill-conditioned. Also: any single-tank W′bal comparator in test 6 must have **its own `τ_W′` fit to the same data**, or the dual-tank wins trivially by having more recovery freedom and the win means nothing.

### 6.8 First empirical results (v0.5) — usefulness, on real rides

Three reviews converged on the point that the paper validated *plausibility* but never the two cheap, data-in-hand questions that decide whether the feature is worth building. Both were run on six real interval/$VO_2$max sessions (the model's best case — a steady endurance ride would score far lower):

- **How often is the PCr bar informative?** The bar carries information beyond single-tank W′bal only when the PCr tank is not full. Across the six rides, **`R_p` is below 95% for ~45% of ride time (range 20–79%)** — well above the ~3% ("curiosity") and ~20% ("real") thresholds a reviewer proposed. On interval sessions the second bar is live nearly half the time. *(Caveat: these are hard interval files; on a steady endurance ride PCr is full most of the time and the figure would be small.)*
- **Does the depletion machinery change the display, vs the recovery-only null model?** Running the full model and the null (§7) on the same traces, the PCr bar diverges by a **mean of ~6 pts but up to ~25–60 pts**, with the maxima at the **starts of recovery valleys** — right after attacks, the decision-relevant window. That is well above the ~5-pt "a rider could act on it" threshold, so **on interval sessions the caps earn their keep**: they set the post-effort PCr level the recovery law then acts on, and a null model would read materially differently exactly when it matters. This is the first evidence that the depletion-side machinery is not merely inert plumbing — though only in the maximal regime (§7 discusses where it is, and is not, doing work).

Neither result validates the *compartments* (that is §6.6, and at the compartment level unfalsifiable in-modality — §6.5, §8). They answer a narrower, real question — *is the two-bar display worth showing, and does the model's complexity change what it shows* — with data instead of assertion, and the answers are yes and yes, for interval training.

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## 7. Assumptions and limitations

- **The PCr/glycolytic split `f_p` is assumed (physiology), and *consistent with* — not corroborated by — the recovery data.** Power cannot identify the depletion split (§4.2). We default to **0.25** from physiological alactic-fraction estimates (~0.20–0.30; di Prampero & Ferretti 1999, Bangsbo 1990 ~20%). The reconstitution curve constrains a `(f_p, τ_g)` ridge, not `f_p`, and physiology is what selects `f_p` on it (§4.1a) — so the two "lines" are not independent, and the recovery data alone would prefer ~0.20. Supported band ~0.20–0.25; a per-athlete target; outputs are sensitive to it.
- **The reserves are a decomposition, not latent state.** The two bars are a physiologically-motivated split of one measured quantity (W′bal), not two independently measured reserves. When the PCr tank is full — which, given `τ_p ≈ 27 s`, is most of the time except the ~30–60 s after a hard effort — the glycolytic bar is an affine transform of the existing single W′bal (`W′bal = f_p + (1−f_p)·R_g/C_g`), so it carries no new information there. The PCr bar adds decision-relevant content **only** in those post-surge transients. Those transients *are* where race decisions concentrate (can I cover this attack right after the last one?), which is the case for the display — but the honest framing

is a heuristic decomposition whose split is assumed, not "the two numbers a rider races on" as if both were measured.

- **The PCr *depletion* rate is also assumed — and it was invisible until v0.4.** The tapered ceiling gives an emergent depletion constant `τ_dep = C_p/P_p_max`, tied to *no* measured PCr depletion kinetic; it is the ratio of three parameters and swings ~3× across plausible riders (~4–13 s). §7 was scrupulous about the assumed *recovery* constants and silent about this equally load-bearing *depletion* one. It should be sanity-checked against literature PCr depletion half-times, not left to fall out of `f_p`, `W′`, and a 1 s sprint power.
- **`τ_g` is calibrated to ~470 (confound resolved); `τ_off` remains assumed.** Ferguson recovered at ~20 W (gate ≈ 0.90), so `τ_g ≈ 470` — not 520, and not the 260 a soft-pedal recovery would have implied (§4.1a). `τ_off = τ_on` is still an assumption (Parolin measured activation, not de-activation; §4.2) and, per Reviewer 2, it is the *only* parameter carrying the depletion-phase novelty: since both bars are affine in W′bal in steady effort, the non-redundant depletion content is the activation-ramp offset, which is a function of `τ_on` (measured) and `τ_off` (not).
- **`LT1` should be measured, not derived from CP.** The `0.80·CP` fallback will be wrong for many riders (LT1 ranges ~65–85% of CP), and it gates whether the glycolytic tank recovers during tempo, so a bad value materially changes recovery estimates.
- **Several small modeling choices are assumptions, flagged not defended.** The LT1 gate shape (`(LT1−P)/LT1`, linear, vanishing just below LT1 — its slope depends on LT1's absolute value, and it arguably switches off metabolite clearance too abruptly through the tempo band); lumping deficit clearance and glycolytic-tank refill onto the *same* `τ_g`; and the spill order (unmet demand spills to glycolytic before PCr). Each is a defensible default, none is derived — named here so a reader weights them accordingly.
- **The deficit `D` corrupts the display precisely at maximal effort.** When `D` is large the two bars read slightly full; `D` grows when power exceeds the (possibly stale) `P_p_max` rate cap, and `P1s` decays intra-ride with fatigue, so this recurs late in hard efforts — a display *failure mode*, not just an accounting footnote. The field should surface a "recalibrate sprint power" hint when `D` grows from full tanks (the `rate_limited` flag; §4.3).
- **The product thesis — two bars → better pacing — is untested as a human-factors claim.** §6.8 shows the bar is *informative* often enough to matter on interval sessions, but not that a rider *decides better* with it. The minimum next step is a usability check (even n=1: race on it, log whether the second bar changed a call) before the display's value is asserted, not just its informativeness.
- **`C_p = 0` means the *usable* alactic store is spent, not that muscle PCr is zero.** Real PCr bottoms out around 20–40% of resting at exhaustion; `C_p` is the usable reserve above that floor, so `R_p = 0` on the display is "usable punch gone," not a muscle state that never occurs.
- **The compartments are unfalsifiable in-modality — permanently, not "not yet."** Two facts compound: the depletion split is unidentifiable from power (§4.2), *and* 31P-MRS cannot be run during real cycling (§6.5). Together these are not "untested" — they are "**no in-modality measurement, even in principle, distinguishes the two-compartment model from a one-compartment model with the same aggregate recovery.**" So for cycling the "two tanks" framing is *terminally* a heuristic: it can be motivated, never confirmed in-modality. The only route to compartment-level evidence is **cross-**

**modality transfer** — measure `τ_p` by knee-extension 31P-MRS (where the magnet works) and assume it transfers — which is itself a nameable, stress-testable assumption, unlike the current no-path-at-all. On *correctness* vs single-tank W′bal there is still no evidence (that needs §6.6). On *usefulness*, v0.5 adds the first real-ride evidence (§6.8): the bar is informative ~45% of interval- ride time and the caps move it materially at decision points — necessary, not sufficient, for the product thesis.

- **Does the depletion-side machinery earn its keep? (Now answered, with data.)** The null model to beat is **a single reserve that depletes exactly as W′bal and recovers bi-exponentially** (fast `τ_p`
- LT1-gated slow `τ_g`, split `f_p`) — the same two bars in recovery, none of the caps/spill/deficit/guards. The answer, from §6.8: the caps **do** move the display on real interval sessions (up to ~25–60 pts at recovery-valley starts), so they are not inert — but they earn their keep in **exactly one regime, the maximal one**, where the tapered ceiling binds and produces the ~23% sprint residual and the post-effort PCr level the recovery law then acts on. Everywhere the ceiling does *not* bind the machinery is inert, and in particular **per-system live consumption (§4.3) is not a reading**: when the ceiling is slack, `take_p = Δ·C_p/(C_p+C_g·g)` contains neither `R_p` nor `R_g` — it settles to a fixed `Δ·f_p` / `Δ·(1−f_p)` split within ~20 s, i.e. it is *(power − CP)* × *a clock*, the power meter rescaled. So: **keep the ceiling (scoped to sprint fidelity), and relabel per-system live consumption as "modelled share," not a measurement** (or drop it). That is the answer to Reviewer 2's S1, drawn from the model's own equations — not "adopt the null wholesale," but "keep the one part that does work and stop claiming the parts that don't."
- **Fixed time constants** ignore the documented pH-dependence and bout-to-bout slowing of PCr recovery unless the optional fatigue term is enabled.
- **Hard CP boundary** omits the aerobic slow component and onset kinetics unless the optional aerobic-ramp term is enabled; expect over-attribution to anaerobic tanks in long low-intensity recovery (the same failure mode reported for the EJAP 2023 model).
- **Power is whole-body and mechanical**, not muscle-specific; the model cannot see local fatigue, cadence, or fiber-type effects.
- **Not medical or diagnostic.** This is a training/pacing aid; tank levels are estimates, not measurements of muscle chemistry.
- **Parameters drift** with fitness, fatigue, heat, and altitude; periodic re-testing is required.

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## 8. Conclusion

The single-tank W′ model earned its place by being power-native and light enough to run anywhere, but it answers "how much anaerobic work is left?" without answering "in which system?" The dual-tank model proposed here offers a physiologically-motivated split at negligible computational cost: divide W′ into a fast, rate-limited, fast-recovering PCr tank and a slow, large, slowly-recovering glycolytic tank; share supra-CP demand between them by **capacity**, cap each by its **peak flux** (PCr tapering with fullness), and refill them with system-specific, intensity-gated laws. It generalizes W′bal in depletion and diverges from it —

deliberately — in recovery, it runs at 1 Hz on a Garmin head unit, and it surfaces a two-system decomposition — *punch left* and *dig left* — that is most informative in the transients where pacing decisions concentrate.

The honest bottom line, stated rather than gestured at: **there is still no evidence the two-bar decomposition is more *correct* than single-tank W′bal** (that needs §6.6, and at the compartment level is unfalsifiable in-modality even in principle — §6.5, §7). What v0.5 *does* add, for the first time, is **usefulness** evidence from real rides (§6.8): on interval sessions the PCr bar is informative ~45% of the time, and — answering the reviewers' null-model challenge with data — the depletion caps move the display by up to ~25–60 pts at the moments that matter, so they are not inert. That resolves the fork more precisely than "ship vs re-architect": **ship the heuristic, keep the ceiling scoped to sprint fidelity, and stop claiming the parts of the machinery the equations show are inert** (per-system live consumption is `(power − CP)` rescaled, not a reading — §7). The `τ_g` confound is closed (Ferguson at 20 W ⇒ 470; §4.1a), and a genuine tension is now hosted rather than buried: Ferguson's *own* recovery components are ~4× slower than the model's while the aggregate fits, and Ferguson concludes the decomposition cannot be done — the strongest single datum in the paper, and it points at the model's premise. What remains decisive is the **recovery-law head-to-head** (§6.6). The physiological framing has run out of room; only that experiment can move this further.

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*Sourcing note:* equations for the incumbent models were cross-checked against PubMed Central full text (PMC7552657, PMC10638188); typeset equations there were image-embedded, so published closed forms were reconstructed from the surviving prose and standard literature. Verify exact coefficients against publisher PDFs before hard-coding.